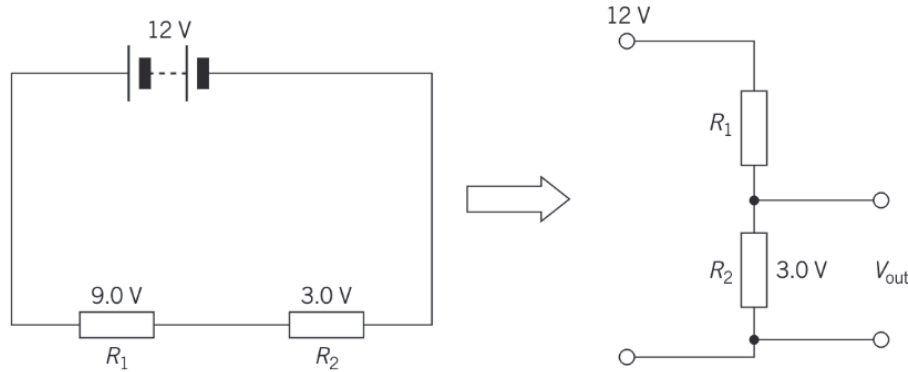


Potential Divider Circuits

Potential Dividers

To divide the p.d. to give any value required up to the maximum supplied by a power source, we can use potential dividers.



A potential divider circuit.

As you can see, the voltage V_{out} is only 3V despite the power supply providing 12V of p.d. because of the **resistance of the resistor**.

Ratio of resistances

The p.d. across each resistor in a potential divider depends on the resistances: the p.d. is shared in the same ratio as the resistances of each resistor.

In other words,

$$R_1 : R_2 : R_3 : \dots : R_n = V_1 : V_2 : V_3 : \dots : V_n$$

and specific to potential dividers,

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

since ratios can be expressed as fractions.

The potential divider equation

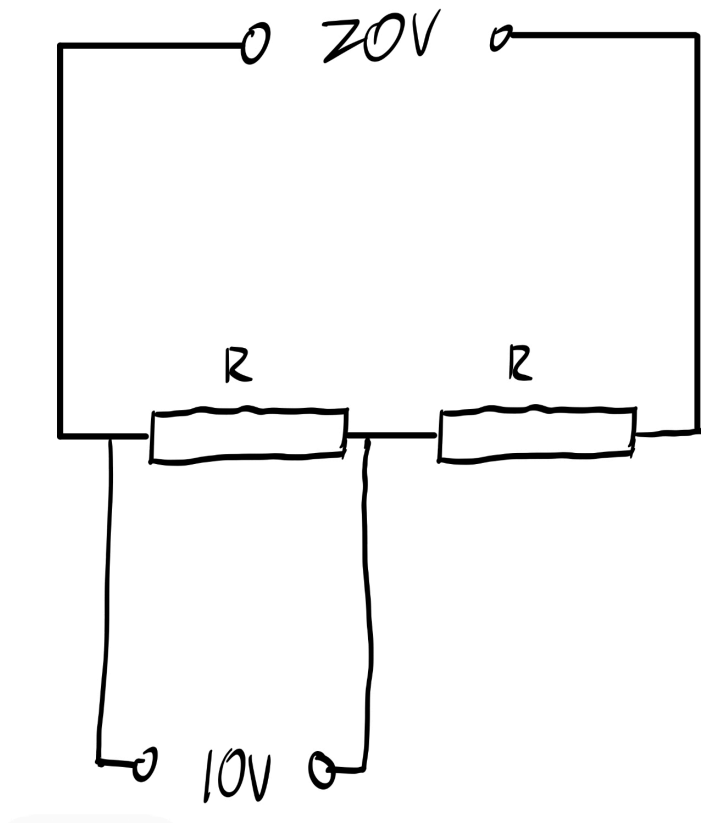
By considering the total p.d. V_{in} and the fraction of the total resistance provided by R_2 the value of V_{out} can be determined by

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{in}$$

This relationship is known as the **potential divider equation**.

Summary Questions

1.

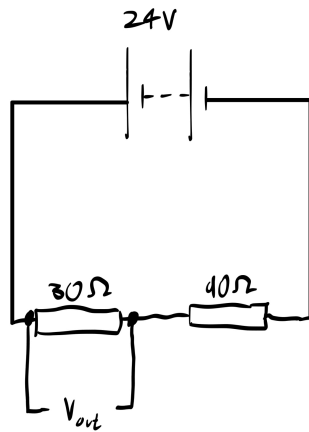


$$\begin{aligned} V_{\text{out}} &= \frac{R_2}{R_1 + R_2} \times V_{\text{in}} \\ &= \frac{R}{2R} \times 20 \\ &= \frac{1}{2} \times 20 \\ &= 10\text{V} \end{aligned}$$

2. a.
$$\begin{aligned} V_{\text{out}} &= \frac{R_2}{R_1 + R_2} \times V_{\text{in}} \\ &= \frac{90}{90 + 270} \times 12 \\ &= 9\text{V} \end{aligned}$$

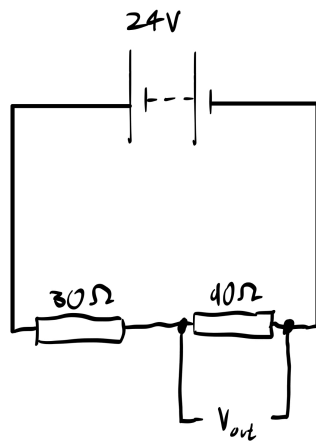
b.
$$\begin{aligned} V_{\text{out}} &= \frac{R_2}{R_1 + R_2} \times V_{\text{in}} \\ &= \frac{120}{120 + 30} \times 60 \\ &= 48\text{V} \end{aligned}$$

3. a.



$$\begin{aligned} V_{out} &= \frac{R_2}{R_1 + R_2} \times V_{in} \\ &= \frac{30}{120} \times 24 \\ &= 6.0V \end{aligned}$$

b.



$$\begin{aligned} V_{out} &= \frac{R_2}{R_1 + R_2} \times V_{in} \\ &= \frac{90}{120} \times 24 \\ &= 18V \end{aligned}$$

4.

$$\begin{aligned} V_{out} &= \frac{R_2}{R_1 + R_2} \times V_{in} \\ \frac{V_{out}}{V_{in}} &= \frac{R_2}{R_1 + R_2} \\ V_{out} R_1 + V_{out} R_2 &= V_{in} R_2 \\ V_{out} R_1 &= V_{in} R_2 - V_{out} R_2 \\ \frac{V_{out} R_1}{V_{in} - V_{out}} &= R_2 \\ \frac{3 \times 110}{360 - 3} &= 0.924\Omega \end{aligned}$$